

Compact-operator conclusion for the unital singleton condition

`MathlibAnnex.Analysis.CStarAlgebra.Representation.faithful_and_compactOperatorModel_of_singleton_amongNonUnital`

theorem

Applies the nonunital compact-operator theorem to the same operator-valued map underlying a unital representation.

Statement

Let A be a nonzero unital complex C^* -algebra, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a unital $*$ -representation satisfying the singleton condition of R11. Then π is injective and

$$\pi(A) = \mathcal{K}(H).$$

Assumptions

The quantified irreducible representations in the singleton condition need not initially be packaged with a unit equation; R11 shows that nonzero irreducibility forces that equation. No separability assumption is made on A .

Conclusion

The same map $a \mapsto \pi(a)$ is a faithful compact-operator model. In particular, it gives a $*$ -isomorphism $A \cong_* \mathcal{K}(H)$.

Proof route

View the same operator-valued function as a representation in the unit-free interface, apply the nonunital singleton theorem, and translate its conclusions back without changing any operator.

▼ Proof steps

1. Let π_0 denote the same function $a \mapsto \pi(a)$, regarded in the representation interface that does not require a unit equation:

$$\pi_0(a) = \pi(a) \quad (a \in A).$$

No operator is changed.

2. The hypothesis of R11 says exactly that π_0 represents the unique unitary-equivalence class of nonzero irreducible $*$ -representations. Apply Faithfulness and compact-operator model from a singleton representation to π_0 .

3. The output is injectivity of π_0 together with $\pi_0(A) = \mathcal{K}(H)$. Since $\pi_0(a) = \pi(a)$ for every a , these are precisely injectivity of π and $\pi(A) = \mathcal{K}(H)$.

Main citations

- Exact theorem and proof
- The singleton condition quantified over all irreducible representations
- The compact-model predicate
- Finite-dimensional representation space
- Surjectivity in finite dimension

Lean source signature (exact)

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theorem faithful_and_compactOperatorModel_of_singleton_amongNonUnital
  [Nontrivial A] [TopologicalSpace.SeparableSpace H]
  (pi : Representation A H)
  (hsingle : IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi) :
  Function.Injective pi ∧
  IsCompactOperatorModel pi.toNonUnitalStarAlgHom
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. Unit structure comes from the surrounding [CStarAlgebra A], not from this condition.
[TopologicalSpace.SeparableSpace H]	The complex Hilbert space H is separable.
(pi : Representation A H)	The specified unital $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$.
(hsingle : IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi)	π is nonzero irreducible and is unitarily equivalent to every nonzero irreducible $*$ -representation of A , even when the input representation interface does not initially record unit preservation.
Function.Injective pi	Faithfulness of the original π .
pi.toNonUnitalStarAlgHom	The same operator-valued map: $\pi_0(a) = \pi(a)$ for every $a \in A$; only the unit equation is omitted from the data interface.
IsCompactOperatorModel pi.toNonUnitalStarAlgHom	The same map is injective and has image exactly $\mathcal{K}(H)$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration:

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